

Tool Survey Proposal - T02 Web Automation Testing

Topic

- **Code:** T02
- **Name:** Web Automation Testing
- **Scope:** End-to-End (E2E) browser tests for the EShop system (including both frontend-web and frontend-admin), focusing strictly on the following flows: Login (FR-02), Add-to-Cart (FR-07), and Checkout (FR-08).

Candidate tools

1. **Playwright:** Traditional automation testing framework.
2. **GitHub Copilot:** AI-augmented tool, used to assist in test generation and test refactoring.
3. **Cypress:** Backup solution.

Comparison matrix

Tool	License cost	Learning curve	EShop fit	AI capability	Community
Playwright	Free, open-source	Medium	Strong: Supports React + Vite, multi-browser testing, features trace viewer and network throttling.	None built-in (requires AI assistance)	Large, growing fast
GitHub Copilot	Paid (trial/student plans available)	Low to medium	Good: When paired with Playwright/Cypress to generate test code and suggest element locators.	High: Code generation and locator suggestions	Very large (GitHub ecosystem)
Cypress	Freemium (Core is OSS)	Low	Good: Solid for web platforms, but primarily optimized for the Chrome-family browsers.	Limited (no native AI integration)	Large, stable

Recommended pick

Playwright (Primary) + Copilot (AI Augmentation)

Rationale

- **Stable and easy to diagnose flakiness:** Features such as *trace-viewer*, *screenshots*, *video*, and *network throttling* are perfect for measuring and resolving flaky test issues across a 10-run execution check.
 - **Fits EShop architecture:** Integrates seamlessly with the React + Vite frontends, smoothly covers the explicit Login/Cart/Checkout flows, and provides excellent multi-browser support.
 - **Boosts productivity:** Copilot speeds up test authoring, generates various assertion types, and suggests locators, while maintaining high reliability through manual human code reviews.
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AI disclosure

- **AI used:** GitHub Copilot (GPT-5.2-Codex model) for initial drafting.
- **Cross-checked:** Thoroughly verified against the T02 brief in [T02_Web_Automation_Testing.docx.md](#) and the EShop Software Requirements Specification (SRS) in the [README.md](#) file. No external web sources were used.